

Fig. 3: An actual-size PCB layout

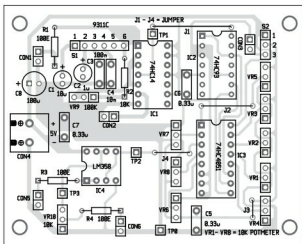


Fig. 4: Component layout of the PCB

Test Points

Test point	Details
TP0	0V
TP1	Square wave output
TP2	Staircase waveform
TP3	Staircase waveform, amplitude depends on position of VR10

by varying these potentiometers. Depending on the selected pins and values of potentiometers, we get different types of staircase waveforms at output pin 3 of IC3. When these waveforms are further given to the input pins of IC4 (dual op-amp), it simply buffers them. Fig. 2 shows four different types of staircase output waveforms obtained at CON5.

Vcc depends on the ICs used in the circuit and can be 5V. With

CMOS ICs like 74Cxxx and CD4000B, we can use Vcc up to 15V.

Construction and testing

An actual-size, single-side PCB layout of the simple adjustable staircase waveform generator is shown in Fig. 3 and its component layout in Fig. 4.

The frequency at pins 1 and 2 of IC1 depends on the type of IC used and values of capacitors and resistors connected to gate N1 of the IC. It can be from 1Hz to 1MHz. If you use an external frequency, move S1 to position 5 to bypass capacitors C1 through C4.

Switch S2 selects the source of reference voltage for VR1 through VR8. In the first position (1), Vcc is used as reference source. In the second position (3), signal from connector CON3 is used as reference

source.

The fixed-amplitude staircase output waveform is obtained at CON5 and variable-amplitude staircase waveform is obtained at CON6.

Each waveform consists of eight clock pulses, zero through seven. The height of each step is adjusted according to the need using potentiometers VR1 through VR8. The frequency of the step is determined by the oscillator around gate N1 of IC1 or by the external clock signal applied to connector CON1. For troubleshooting, check the various test points and ensure you get the outputs as per table. ●

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